DEPARTMENT OF MATHEMATICS AND APPLIED MATHEMATICS

Mathematics II Advanced Calculus (2AC)

Tutorial Sheet 9: Solutions **20th and 21st April 2023**

1. Say whether the following statement is true or false. Explain clearly your answer.

   Let $f : \mathbb{R}^2 \to \mathbb{R}$ be a function with continuous partial derivatives up to order two. If a critical point (a, b) of the function f is such that

   $$f_{xx}(a, b) > 0 \quad \text{and} \quad f_{xx}(a, b) + f_{yy}(a, b) = 0\,,$$

   then the point (a, b) is a point of local minimum for the function f.

   Using $f_{yy}(a, b) = -f_{xx}(a, b)$, we get that

   $$\begin{aligned} D(a,b) &= f_{xx}(a,b)f_{yy}(a,b) - \big[f_{xy}(a,b)\big]^2 = -\big[f_{xx}(a,b)\big]^2 - \big[f_{xy}(a,b)\big]^2 \\ &= -\Big(\big[f_{xx}(a,b)\big]^2 + \big[f_{xy}(a,b)\big]^2\Big) < 0 \quad \textbf{(Why can't we have } D(a,b) = 0 \textbf{ here!)}\,. \end{aligned}$$

   So, (a, b) is a saddle point and therefore, the statement is **false!**

2. Given the function $f(x, y) := xy + x - y$.

   (i) The only critical point is $(1, -1)$. The determinant of the hessian matrix is $-1 < 0$. Therefore $(1, -1)$ is a saddle point.

   (ii) Use Lagrange multipliers we find

   $$\begin{aligned} &y + 1 - 2\lambda x = 0 \\ &x - 1 - 2\lambda y = 0 \qquad (*) \\ &x^2 + y^2 - 25 = 0 \end{aligned}$$

   First solution: from the first equation we get $y = 2\lambda x - 1$ that we substitute if the second equation and get $x(1 - 4\lambda^2) = 1 - 2\lambda$. Now, to solve this equation to find x, we need to check whether $1 - 4\lambda^2 \neq 0$ or $1 - 4\lambda^2 = 0$.

   (a) The case $1 - 4\lambda^2 = 0$, that is $\lambda = \pm\frac{1}{2}$: we get for $\lambda = -\frac{1}{2}$ that $0x = 2$ which is impossible. For $\lambda = \frac{1}{2}$ we get $0x = 0$ and we go back to the system $(*)$ which becomes

   $$\begin{cases} y = x - 1, \\ x^2 + y^2 = 25 \end{cases} \Rightarrow \quad x^2 + (x-1)^2 = 25 \quad \Leftrightarrow \quad x^2 - x - 12 = 0 \quad x = -3,\, 4\,.$$

   Thus, we find the points $\boxed{(-3, -4) \quad \text{and} \quad (4, 3)}$.

(b) The case $1-4\lambda^2 \neq 0$, that is, $\lambda \neq \pm\frac{1}{2}$. We find that

$$x = \frac{1-2\lambda}{1-4\lambda^2} = \frac{1-2\lambda}{(1-2\lambda)(1+2\lambda)} = \frac{1}{1+2\lambda}$$

and $y = -\dfrac{1}{1+2\lambda}$ which imply that $x = -y$. So, from the last equation in $(*)$ we get $2x^2 = 25$ that is, $x = \pm\dfrac{5}{\sqrt{2}}$. Hence, we get the points $\boxed{(\frac{5}{\sqrt{2}}, -\frac{5}{\sqrt{2}}) \quad \text{and} \quad (-\frac{5}{\sqrt{2}}, \frac{5}{\sqrt{2}})}$.

Now, find the values of f at the points in (a) and (b); the smallest of these values is the minimum value of f on the circle $x^2+y^2 = 25$ while the largest is the maximum.

Second solution: (from a student!) From the system

$$\begin{aligned} y+1-2\lambda x &= 0 \\ x-1-2\lambda y &= 0 \qquad (*) \\ x^2+y^2-25 &= 0 \end{aligned}$$

we add-up the first two equations to get

$$y+x-2\lambda(x+y) = 0 \quad \Leftrightarrow \quad (1-2\lambda)(x+y) = 0 \quad \Leftrightarrow \quad 1-2\lambda = 0 \text{ or } x+y = 0\,.$$

Hence, we simply have the two cases $\lambda = \dfrac{1}{2}$ which gives the points $(-3,-4)$ and $(4,3)$ as in the first solution and the case $y = -x$ from which we get the points $(\frac{5}{\sqrt{2}}, -\frac{5}{\sqrt{2}})$ and $(-\frac{5}{\sqrt{2}}, \frac{5}{\sqrt{2}})$ also as above.

Now, we evaluate the function f at those four points and decide which ones are points of maximum and which ones are points of minimum.

3. Find the points of absolute maximum and minimum of the function $f(x,y) := -x+y^2$ over the ellipse $E := \{(x,y) \in \mathbb{R}^2\colon\, x^2+4y^2 = 1\}$.

We use the method of Lagrange multipliers. We seek λ, x, y such that

$$\begin{cases} -1-2\lambda x = 0 \\ 2y-8y\lambda = 0 \\ x^2+4y^2 = 1 \end{cases}$$

From the second equation we find $y = 0$ or $\lambda = \dfrac{1}{4}$.
For $y = 0$, we find from the third equation $x = \pm 1$ and from these values we can find the corresponding values of λ from the first equatiomn. Hence, we get the points $(-1,0)$ and $(1,0)$.
For $\lambda = \frac{1}{4}$, we get from the first equation that $x = -2$ and from the third equation that $3+4y^2 = 0$ which is impossible. Therefore, we have two points $\boxed{(-1,0) \quad \text{and} \quad (1,0)}$.
Since $f(-1,0) = 1$ and $f(1,0) = -1$, we get that $(-1,0)$ is the point of absolute maximum while $(1,0)$ is the point of absolute minimum.

4. Consider the function $f(x,y,z) = x+y+z$. Find the maximum and minimum values of f on the sphere $x^2+y^2+z^2=1$.
We set $g(x,y,z) = x^2+y^2+z^2-1$ and hence, we first find x, y, z, λ such that

$$\begin{cases} f_x(x,y,z)) = \lambda g_x(x,y,z) \\ f_y(x,y,z) = \lambda g_y(x,y,z) \\ f_z(x,y,z) = \lambda g_z(x,y,z) \\ g(x,y,z) = 0 \end{cases} \Leftrightarrow \begin{cases} 1 = 2\lambda x \\ 1 = 2\lambda y \\ 1 = 2\lambda z \\ x^2+y^2+z^2 = 1. \end{cases}$$

From the first three equations, we find that $x = y = z = \dfrac{1}{2\lambda}$. Now substituting in the last equation, we get $\lambda = \pm\dfrac{\sqrt{3}}{2}$ and hence, the points $(-\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}})$ and $(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}})$. $f(-\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}) = -\sqrt{3}$ is the minimum while $f(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}) = \sqrt{3}$ is the maximum.

5. A box has to be constructed out of 120 cm^2 of material. Find the dimensions x, y and z of the box that maximize the volume.

We want to solve the problem $\begin{cases} \text{Maximize } V(x,y,z) = xyz \\ \text{subject to } 2xy+2xz+2yz = 120\,. \end{cases}$
where $2xy+2xz+2yz$ represents the total area of the 6 faces of the box.

(a) **First solution:** Using the method of the Lagrange multiplier, we set

$$A(x,y,z) := xy+xz+yz-60$$

and we seek x, y, z, λ such that

$$\underbrace{\begin{cases} V_x(x,y,z) = \lambda A_x(x,y,z) \\ V_y(x,y,z) = \lambda A_y(x,y,z) \\ V_z(x,y,z) = \lambda A_z(x,y,z) \\ A(x,y,z) = 0, \end{cases}}_{(*)} \Leftrightarrow \underbrace{\begin{cases} yz = \lambda(y+z) \\ xz = \lambda(x+z), \\ xy = \lambda(x+y), \\ xy+xz+yz-60 = 0, \end{cases}}_{(**)} \Leftrightarrow \underbrace{\begin{cases} xyz = \lambda x(y+z) \\ xyz = \lambda y(x+z), \\ xyz = \lambda z(x+y), \\ xy+xz+yz-60 = 0, \end{cases}}_{(***)}$$

where you obtained the last system $(***)$ by multpliying the first three equations by x, y and z respectively in the system $(**)$. So, we get from the first three equations in $(***)$ that

$$\underbrace{\lambda x(y+z) = \lambda y(x+z) = \lambda z(x+y)}_{\lambda\neq 0} \quad\Leftrightarrow\quad x(y+z) = y(x+z) = z(x+y)\,.$$

Thus we get

$$\begin{cases} x(y+z) = y(x+z) \\ y(x+z) = z(x+y) \end{cases} \Leftrightarrow \begin{cases} xz = yz \\ yx = zx \end{cases} \Leftrightarrow \boxed{x = y = z} \text{ (since } x>0,\ y>0,\ z>0\,.$$

So, if we substitute $x = y = z$ into the fourth equation in $(***)$, we get

$$3x^2-60 \quad\Leftrightarrow\quad x^2 = 20 \quad\Leftrightarrow\quad x = \sqrt{20} = y = z \quad\Rightarrow\quad \lambda = \frac{x^2}{2x} = \frac{20}{2\sqrt{20}} = \frac{\sqrt{20}}{2} = \sqrt{5}\,.$$

Hence, the dimensions x, y and z of the box that maximize the volume are $x = y = z = \sqrt{20}$. **Notice that these dimensions can only realize the maximum since the minimum value here cannot be not achieved!**

(b) **Second solution:** Solve z from the constraint, solve $z = \dfrac{60 - xy}{x+y}$, substitute into the expression of V and get the function of two variables

$$f(x,y) = \frac{xy(60-xy)}{x+y}$$

for which we find the maximum value in the set $\{(x,y)\colon\, x > 0,\, y > 0,\, xy < 60\}$. Let us find the critical points of f.

$$\begin{cases} \dfrac{\partial f}{\partial x} = \dfrac{y^2(60 - x^2 - 2xy)}{(x+y)^2} = 0 \\[2ex] \dfrac{\partial f}{\partial y} = \dfrac{x^2(60 - y^2 - 2xy)}{(x+y)^2} = 0 \end{cases} \Rightarrow \begin{cases} 60 - x^2 - 2xy = 0 \\ 60 - y^2 - 2xy = 0 \end{cases}$$

Subtracting the two equations we get $x^2 - y^2 = 0$ which implies $x = y$. Substituting in the first equation above we obtain $60 - 3x^2 = 0$ which implies that $x = y = \sqrt{20}$. Hence, also $z = \sqrt{20}$. So, the box of maximum volume is a cube with size $\sqrt{20}$.

6. **(After Max/min-IV to be covered in the lecture on Friday)** Given the function $f(x,y) := xy + x - y$. Let $D := \{(x,y) \in \mathbb{R}^2\colon\, x^2 + y^2 \le 25 \text{ and } x \ge 0\}$. Find the points of maximum and minimum of f on D.

Following the theory in the lecture notes Max/Min-IV, we split the problem into two: finding the points of maximum and minimum in the interior of the domain D and then on its boundary. The points of maximum and minimum in the interior are critical points of the function f.

(i) We first find the critical points of f lying in the interior of the domain D. The only critical point is $(1,-1)$ which lies indeed in the interior of D. In fact,

$$1^2 + (-1)^2 = 2 < 25 \quad \text{and} \quad 1 > 0\,.$$

(ii) Let us now find the maximum and minimum of f on the boundary of D. Now, the boundary of the domain D has two pieces: The right-half circle $x^2 + y^2 = 25$ with $x \ge 0$ and the segment line L on the y-axis joining the points $(0,-5)$ and $(0,5)$.

Let us first find the points of maximum and minimum of the function f over the right-half circle through the method the Lagrange multiplier. So we seek x, y, λ such that

$$(*) \qquad \begin{cases} y + 1 = 2\lambda x, \\ x - 1 = 2\lambda y, \\ x^2 + y^2 = 25, \quad x \ge 0\,. \end{cases}$$

So, we get from Question 2, the only points $(5/\sqrt{2}, -5/\sqrt{2})$ and $(4,3)$ which correspond to $x \ge 0$.

(ii) Now, we find the points of maximum and minimum of the function f over the segment line L which equivalent to finding the points of maximum and minimum of the function $f(0,y) = -y$ for $y \in [-5,5]$. We get the points $(0,-5)$ and $(0,5)$.

(iv) We find that

$$f(1,-1) = 1; f(5/\sqrt{2}, -5/\sqrt{2}) = -\frac{25}{2} + \frac{5}{\sqrt{2}} + \frac{5}{\sqrt{2}} = -\frac{25}{2} + 5\sqrt{2} \approx -5,83\,.$$
$$f(4,3) = 12 + 3 - 4 = 11, \quad f(0,-5) = 5\,, \quad f(0,5) = -5\,.$$

Hence, the function f has maximum value on D equal to 13 which is achieved at the point $(4,3)$ and minimum value equal to $\dfrac{10\sqrt{2}-25}{2}$ achieved at the point $(5/\sqrt{2}, -5/\sqrt{2})$.